

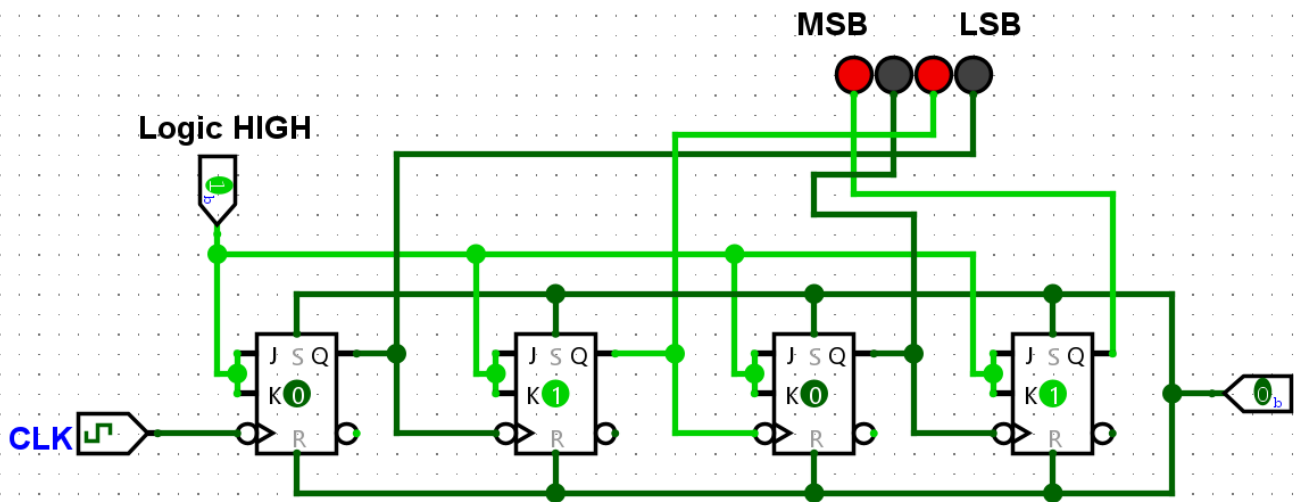


Figure 1: JK Flip Flop Implementations of a 4-bit Asynchronous Up-Counter

```

1 `timescale 1ns/1ps
2
3 // =====
4 // T Flip-Flop - Negative edge triggered, toggle mode
5 // =====
6 module tff (
7     input clk,
8     input reset, // active HIGH async reset
9     output reg q
10 );
11
12 always @(negedge clk or posedge reset) begin
13     if (reset) q <= 0;
14     else q <= ~q;
15 end
16 endmodule
17
18 // =====
19 // 4-bit Asynchronous Ripple Counter
20 // Each FF output drives the clock of the next - ripple effect
21 // Equivalent to 7493 IC configuration
22 // =====
23 module async_counter_4bit (
24     input clk,
25     input reset,
26     output wire [3:0] q
27 );
28
29 // QA clocked by external clk
30 // QB clocked by QA
31 // QC clocked by QB
32 // QD clocked by QC
33 tff ff0 (.clk(clk), .reset(reset), .q(q[0]));
34 tff ff1 (.clk(q[0]), .reset(reset), .q(q[1]));
35 tff ff2 (.clk(q[1]), .reset(reset), .q(q[2]));
36 tff ff3 (.clk(q[2]), .reset(reset), .q(q[3]));
37 endmodule

```

Figure 2: Verilog script and Truth table Implementations of a 4-bit Counter

```

1 `timescale 1ns/1ps
2
3 module tb_async_counter;
4
5     reg clk, reset;
6     wire [3:0] q;
7
8     async_counter_4bit uut (
9         .clk (clk),
10        .reset (reset),
11        .q (q)
12    );
13
14    // Clock: 20ns period
15    initial clk = 0;
16    always #10 clk = ~clk;
17
18    // VCD dump - dumps all internal FF signals too
19    initial begin
20        $dumpfile("async_counter.vcd");
21        $dumpvars(0, tb_async_counter);
22    end
23
24    initial begin
25        // Reset
26        reset = 1; #25;
27        reset = 0;
28
29        $display("");
30        $display(" Clock Pulse | QD QC QB QA | Decimal");
31        $display("-----");
32
33        repeat(17) begin
34            @(negedge clk); #2;
35            $display(" | %b %b %b %b | %0d",
36                q[3], q[2], q[1], q[0], q);
37        end
38
39        $display("-----");
40        $display("");
41
42        // Run 2 full cycles for clean GTKWave view
43        repeat(32) @(negedge clk);
44
45        $display("Done. Open async_counter.vcd in GTKWave.");
46        $finish;
47    end
48 endmodule

```

QD	QC	QB	QA	Decimal
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15
0	0	0	0	0
0	0	0	1	1

for the T-type Flip Flop Asynchronous Up-

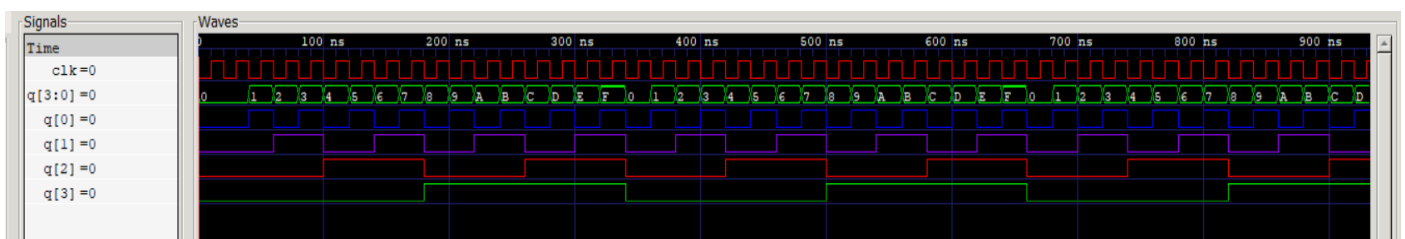


Figure 3: Signal Response for the T-type Flip Flop Implementations of a 4-bit Asynchronous Up-Counter